Chapter 1

The Normal Distribution

The Normal distribution, also known as the Gaussian distribution, is a type of continuous probability distribution for a real-valued random variable. It is one of the most important probability distributions in statistics due to its several unique properties and usefulness in many areas.

1.1 Defining the Normal Distribution

The Normal distribution is defined by its mean (μ) and standard deviation (σ). The probability density function (pdf) of a Normal distribution is given by:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where:

- x is the point up to which the function is integrated,
- μ is the mean or expectation of the distribution,
- σ is the standard deviation,
- σ^2 is the variance.

1.2 Importance of the Normal Distribution

There are several reasons why the Normal distribution is crucial in statistics:

- **Central Limit Theorem:** One of the main reasons for the importance of the Normal distribution is the Central Limit Theorem (CLT). The CLT states that the distribution of the sum (or average) of a large number of independent, identically distributed variables approaches a Normal distribution, regardless of the shape of the original distribution.
- **Symmetry:** The Normal distribution is symmetric, which simplifies both the theoretical analysis and the interpretation of statistical results.

- **Characterized by Two Parameters:** The Normal distribution is fully characterized by its mean and standard deviation. The mean determines the center of the distribution, and the standard deviation determines the spread or girth of the distribution.
- **Common in Nature:** Many natural phenomena follow a Normal distribution. This includes characteristics like people's heights or IQ scores, measurement errors in experiments, and many others.

Given its properties, the Normal distribution serves as a foundation for many statistical procedures and concepts, including hypothesis testing, confidence intervals, and linear regression analysis.

1.2.1 Example

A set of course grades is modeled by a normal distribution with mean $\mu = 75$ and standard deviation $\sigma = 8$. Using a 90–80–70–60 grading scale, find the percentage of students who earned a B (scores from 80 to less than 90) using Excel and Python.

Solution

Let X be a student's grade. Then

$$X \sim N(75, 8).$$

A B corresponds to the interval $80 \leq X < 90$, so we want

$$P(80 \leq X < 90).$$

Excel computes cumulative probabilities using

```
NORM.DIST(x, mean, standard_dev, TRUE)
```

Therefore,

$$P(80 \leq X < 90) = P(X < 90) - P(X < 80).$$

In Excel, enter:

```
=NORM.DIST(90,75,8,TRUE) - NORM.DIST(80,75,8,TRUE)
```

Excel returns approximately

$$0.2356.$$

So about 23.56% of students earned a B.

Now compute the same probability in Python.

```
from scipy.stats import norm

mu = 75
sigma = 8

probability = norm.cdf(90, mu, sigma) - norm.cdf(80, mu, sigma)

print(probability)
print(probability * 100)
```

Python returns approximately

$$0.2356$$

which confirms the Excel result.

Both Excel and Python are computing the same cumulative distribution function.

Exercise 1

Working Space

A batch of parts has diameters modeled by a normal distribution with mean $\mu = 10$ mm and standard deviation $\sigma = 2$ mm:

$$X \sim N(10, 2).$$

A part is considered *acceptable* if its diameter is between 6.72 mm and 12.28 mm.

(a) Write the probability statement that represents the fraction of acceptable parts.

(b) Use Excel to compute this probability using `NORM.DIST`.

(c) Convert your probability to a percentage.

Answer on Page 7

APPENDIX A

Answers to Exercises

Answer to Exercise 1 (on page 4)

(a) The fraction of acceptable parts is

$$P(6.72 \leq X \leq 12.28).$$

(b) Since `NORM.DIST(x, mean, sd, TRUE)` returns $P(X \leq x)$, we compute

$$P(6.72 \leq X \leq 12.28) = P(X \leq 12.28) - P(X \leq 6.72).$$

In Excel, enter:

```
=NORM.DIST(12.28,10,2,TRUE) - NORM.DIST(6.72,10,2,TRUE).
```

(c) Excel returns approximately 0.8224, so the percentage acceptable is

$$82.24\%.$$

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